

Abdelwahab Mahmoud

Senior Software Engineer · Full-stack & Platform

Cairo, Egypt

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Summary

Senior Software / Full-Stack Platform Engineer with 8+ years building and operating AWS production systems: Node.js/TypeScript services, a delivery platform serving 100K daily users, and WPML's LLM translation and agent platform on 1.5M+ WordPress sites. Across the stack: ECS Fargate, Terraform, SQS, CI/CD, Datadog and OpenTelemetry for observability, Aurora, Redis and React. I build the feature, own the platform from architecture through deployment to reliability. Usually the one who finds out why production broke.

Work Experience

OnTheGoSystems (WPML — WordPress plugins)

Remote (Hong Kong)

Senior Software Engineer

Dec 2024 - Present

- Built the multi-provider translation LLM engine behind WPML (Bedrock, OpenAI, Gemini): fallback chains, JSON repair and rate-limit retries keep translations flowing when a provider degrades.
- Built the guardrails agents run behind: an MCP server exposing live databases, logs and selected APIs through authenticated, monitored access, adopted as the team's standard agent integration path.
- Owned credits and billing correctness for a metered LLM product; authorization architecture replacing scattered permission checks.
- Own an agentic triage platform handling first-line incident and support inquiries, cutting Mean Time To Resolve.
- Cleared Aurora MySQL's worst-query backlog query by query: rewritten plans, covering indexes and restructured hot reads, until CPU pressure stopped turning into incidents.
- Replaced ECS autoscaling with a queue-depth Lambda autoscaler that drains long-running jobs instead of killing them.
- Owned the GitLab CI/CD pipelines: build, e2e test, SAST and deploy gates for every service in the product.
- Kept plugin APIs backwards compatible across releases with deprecation paths and contract tests.
- Migrated the product onto AWS ECS Fargate running Docker containers, and own the platform and its Terraform.

Dailymealz

Riyadh, Saudi Arabia

Senior Software Engineer

Dec 2021 - Dec 2024

- Built the real-time order service behind a 100,000 daily-user delivery product: kitchen, driver and delivery status pushed to mobile and web clients over Node.js and Socket.IO.
- Cut Mean Time To Resolve by 30% by instrumenting order, driver and kitchen flows with Datadog and OpenTelemetry.
- Decomposed the fulfillment component out of the PHP/Laravel monolith into a standalone service in a distributed event-driven architecture, passing state changes over AWS SQS behind a REST API contract.
- Chose incremental extraction over a full event-driven rewrite: a team of six couldn't freeze features for months to run two partially consistent systems.
- Served hot order state from Redis in a write-behind cache pattern with MySQL behind it, keeping status fan-out off the transactional path.
- Migrated the real-time order service from JavaScript to TypeScript so event and message payload mismatches fail at compile time, not at runtime in front of customers.
- Led a team of 5 building a driver location check-in system: planning, reviews and pairing across the driver app.

Nformacy

Cairo, Egypt

Full-Stack Developer

Oct 2020 - Dec 2021

- Led full-stack features from requirements to deployment, and moved the web apps to a SaaS model.
- Built mentor calendar booking with Zoom, ClickUp and calendar integrations.
- Wrote BDD tests and managed CI/CD pipelines.

DevSquads (formerly Fikr Labs)

Cairo, Egypt

Software Engineer

Jan 2018 - Oct 2020

- Built software in self-organizing Extreme Programming teams with TDD, thin vertical slices and test-covered refactoring of legacy code, across React Native, Ruby on Rails, Java Spring and React.
- Mentored teams adopting XP and Agile practices: pairing, code review discipline, iterative planning.

Skills

Programming Languages

Backend Development

Frontend Development

AI & LLM

Cloud & Infrastructure

Observability

Databases

Testing

WordPress & Compatibility

Spoken Languages

PHP, TypeScript, JavaScript, Go, Ruby, Java (Spring), Python, SQL

Ruby on Rails, Node.js, REST API design, service-oriented architecture, SQS, Kafka

Next.js, React, Redux, Hotwire (Turbo & Stimulus), React Native, Tailwind CSS

AWS Bedrock, OpenAI, Gemini, agentic workflows, Claude Agent SDK, MCP, Langfuse, evals

AWS (ECS Fargate, Lambda, SQS, Step Functions, CloudWatch, S3, IAM), Kubernetes,

Terraform, Docker, GitHub Actions, GitLab CI

Datadog, OpenTelemetry, AWS CloudWatch, Langfuse tracing

PostgreSQL, MySQL, Aurora, Redis, DynamoDB, ClickHouse

TDD, Spec-driven development, RSpec, Capybara, Playwright, Cypress, Jest, VCR

WPML ecosystem, plugin update safety, backwards compatibility, i18n

English, Arabic (native)

Open Source Projects

ecs-drain-scaler

<https://github.com/Abdelwahab313/ecs-drain-scaler>

2026

- **AWS (ECS, SQS), Terraform, Go** — backlog-driven scaling for queue workers on ECS Fargate plus a drain protocol so scale-in can't kill a task mid-job. Scaling policy implemented with tests; design decisions written up as ADRs. Apache-2.0.

focusflow

<https://github.com/Abdelwahab313/focusflow>

2026

- **TypeScript, PocketBase, GitHub Actions** — offline-first daily planner and focus timer: plan against a backlog, run focus sessions on one task, review planned vs actual time. Optional sync to a self-hosted PocketBase. MIT.

handoff-mcp-server

<https://github.com/Abdelwahab313/handoff-mcp-server>

2026

- **TypeScript, MCP** — MCP server that saves and restores an AI coding agent's task context as a validated briefing between sessions.

egy-finance-radar

<https://github.com/Abdelwahab313/egy-finance-radar>

2026

- **Python, PostgreSQL, Next.js** — EGX research tool: fundamentals classification, technical signals and dated recommendations behind a dashboard.

Education

Faculty of Science Cairo University

Bachelor of Science in Geophysics

Cairo, Egypt

Sep 2012 - Jun 2016